

SCRIPT –

Good morning, everyone, glad to be presenting our research paper titled Hierarchical optimization technique for placement of visual landmarks for robot navigation in repetitive structured indoor environments. First we'll look into abstract move to the introduction, identify the research gaps and produce novelty of the work setup objectives, followed by the methodology and results.

Slide: abstract

I'm pretty sure we have all seen robots in a hotel navigating indoors carrying things from one place to another, the trend for autonomous service robots has become more integrated into our daily lives than ever. In environments like these like the hotel corridors, hospitals, malls, robots are generally denied of GPS access and wireless signals are unreliable due to multipath effects and interferences. Oh using slam might be the solution, but no this is not exactly efficient for repetitive indoor geometries which are identical for the most part, which creates severe ambiguity for visual localization.

Slide: Introduction

To resolve the spatial ambiguity, we utilize visual markers, specifically ArUco tags as navigation beacons. These ArUco tags are low cost, computationally lightweight and easily detectable in real time. Now that the solution for the problem is figured out, there comes the next question, how is this method effective, efficient and reliable, all of these questions can be answered by asking one more question, how many of these markers to be placed and where exactly solves the puzzle, and this is the is exactly the core discussion of the paper.

Slide: Research Gap and Novelty

Existing research in the field were studied in detail, all the papers studied discussed that placing the markers uniformly is not efficient and especially across the turns, there is a notable localization failure. The papers has discussed geometric constraints-based placement or feature based placements to be a solution for this problem.

Our work introduces a novel scalable alternative of combining the geometric based constraint placement, visibility aware strategies and feature based placement stacked in the form of layers hieratically to form the multi layered hierarchal optimization technique for placement of visual landmarks.

Slide: Objectives

The objectives of this research is, First to calibrate the onboard camera system to determine the optimal size and ensure accurate detection of ArUco markers. Second is to to configure functionally specific markers and conduct a mathematical feasibility

analysis for marker placement. Third is to design the hierarchical optimization technique by analyzing spatial constraints, visibility requirements, and navigation performance. And finally validate our proposed technique utilizing both simulation software and a real-time hardware environment.

Slide: Methodology

The work started by defining the test environment, that aligns with the problem statement featuring repetitive environments, functional door areas and junction zones, which are prone to localization errors. The geometry is mapped into MATLAB, Webots simulation software and a scaled physical environment.

Slide: Manual placement

The next step was to place the markers manually, according to the paper's studies, markers was setup uniformly from a constant distance from each other. And on the other hand to understand geometric constraint-based placement, the camera parameters such as focal length was used to obtain the maximum detection distance a tag must be placed and the maximum horizontal field of view and place the tags according to the visibility constraints. Three different layouts were drafted to understand the effective placement, one was to place all the tags on single side, the second one was to place the tags on alternate side of the walls, and random placement layout. All of the tags were placed manually after calculating the visibility constraints.

Slide: setup

The result obtained showcased that the optimal visibility constraints based placement on random layout performed better, and atleast one tag was visible at 89% time of the whole run. While it can also be seen that there is a lack of availability of tags in junctions

Slide: Inference

Moving on, the placement strategies and the inferences was collected to draft the algorithm. These were the main points that were noted. The first one is the marker design. An optimal marker size is important, small markers resulted in poor detection whereas oversized markers created unwanted redundancy and congestion in readings, Therefore a balance is required, The second inference generated was based on the layout designing, uniform placement resulted in high redundancy, and also increased infrastructure cost whilst still with low localization availability, whereas in the visibility based constraint design, it was noted that alternatively placing the tags on walls was effective for wide corridors and single side was sufficient for narrow corridors. The last inference generated showcased the importance for the need of feature-based placement, as noted from the results, it could be seen that the junction availability was at 75% where the localization errors are prone to occur. So a feature-based placement is needed.

Slide: Hierarchical optimization framework

This is the stage where we start drafting the algorithm, problem statement for the algorithm is to ensure reliable indoor robot navigation with minimal infrastructure. Minimal infrastructure refers to the trade-off between reduction of Aruco tags while increasing the localization availability.

Slide: Feasibility analysis

The first step of the algorithm is the Feasibility analysis, a set of tag sizes with pixel thresholds are given along with the environment geometries like the corridor length, width. Using the formulae shared, the maximum detection distance, and maximum horizontal field of view is obtained as inputs for the tag placement. The values are stored and a feasibility analysis of the tag size is done, based on the detection distance and the corridor space availability, the effective tag size is chosen. The tag size is stored and the tags with the chosen size are placed next to each other in a uniform arrangement as shown in the figure and the candidate locations are stored.

Slide: Visibility Matrix

The next step is to utilize the stored inputs, and derive a visibility constraint-based matrix of chosen tag candidates, that will provide an efficient localization availability wrt to the lateral visibility during turns and maximum corridor coverage. This is the condition of the visibility matrix and only a few set of tag candidates are passed for the next step, out of the 178 tags, now 164 tags are chosen for next step.

Slide: Multi stage Tag placement

This step features the feature based placement of tags as noted from the third inference the importance of feature based placement, the tags are divided into three categories based on importance ranks, the first in the list is functional tags, the effective guide to say that the robot has reached the location or must start from this location, example are the tags placed at doors or task areas, The second on the list is the junction based placement, to reduce the localization failures in the error prone areas these two categories are important and tags must be present mandatorily. The last in the list is the corridor tags, since corridors are straight layouts, the localization failures is minimal.

Slide: Results Iterative optimization

To fill the remaining corridor gaps, we utilize a greedy iterative search algorithm. At each step, the algorithm selects a candidate tag that maximizes new pose coverage. Crucially, it applies a strict penalty score to any tag placed at close range on the same wall as an existing tag, effectively eliminating unnecessary redundancy. Over 12 iterations, the algorithm systematically reduced coverage gaps until reaching a stable, optimal configuration

Slide: Evaluation Report

The results clearly highlight the efficiency of this approach. Out of an initial pool of 178 candidate tags, our algorithm finalized a layout utilizing only 32 tags. This represents an 82 percent reduction in physical infrastructure. Despite this massive reduction, we achieved a total localization availability of 94.2 percent and minimized the total blackout length to a mere 0.3 meters.

Slide: Simulation environment

To ensure these mathematical models translate to dynamic environments, we validated the optimized coordinates in the Webots robotics simulator. Operating over a approx 14-meter path, the simulated robot achieved a mean visibility of 1.011 tags per frame, a failure rate of just 6.32 percent, and a low redundancy rate of 8.21 percent

Slide: Hardware utilized

Finally, we performed real-world hardware validation. We deployed a Yahboom Jetbot, powered by an NVIDIA Jetson Nano and an 8-megapixel Sony IMX219 camera featuring a 160-degree ultra-wide field of view, inside a physical corridor.

Slide: Validation in real world

The real-world metrics strongly correlated with our simulations. We captured 2,666 frames across various speed profiles ranging from slow to high speeds achieving a localization availability of 91.82 percent and an incredibly low blackout rate of 8.18 percent. Most importantly, we achieved a reliable 100 percent marker availability at all junctions, completely resolving the critical issue of turning blind spots

Slide: Conclusion

In conclusion, our visibility-aware hierarchical optimization framework provides a practical, automated solution to the complex problem of visual beacon placement in repetitive environments. By systematically analyzing feasibility, prioritizing structural necessities like junctions and doors, and applying greedy optimization for the corridors, we successfully minimized infrastructure costs while maintaining robust, deterministic navigation.

As we look to the future, we plan to expand this work by developing an end-to-end automated placement strategy utilizing machine learning techniques. Additionally, we will investigate sensor fusion, combining visual marker data with IMU and wheel odometry, to further bridge the gap between low-cost hardware and high-precision indoor autonomy

